# Pavel Vasilyev

## Senior AI Engineer - LLM Systems, APIs & Applied ML

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### PROFILE

Senior AI engineer with fifteen years across applied ML, NLP, and software. I build LLM systems where routing, fallback, cost, and latency are explicit and measurable, and I design the APIs and data flows around them.

### FIT FOR THE ROLE

- Multi-step LLM workflows across Claude, Gemini, OpenRouter, Ollama, and Cloudflare Workers AI: provider routing, role-specific fallback chains, retry ceilings, time and token budgets, bounded concurrency, and per-call token and latency tracing
- Classical ML and NLP depth: feature engineering, model selection and training, text classification and extraction
- Clean API and service design: authenticated HTTP APIs, credential handling, Phoenix services deployed behind Caddy and systemd
- Working with product and design: turning ambiguous requests into written specifications and explicit trade-off decisions

### EXPERIENCE

**Independent Software Practice** - consulting and open source 2010-present

Consulting, advisory, and software practice between salaried roles.

- Built tools and technical systems against client specifications while maintaining an open-source body of 150+ projects across eight or more languages.

**Research Lead** - Spring Research 2020-2022

Quantitative research firm focused on financial-market modelling.

- Owned the full applied-research cycle: monitored financial markets, analysed data, developed new signal representations and transformations, extracted ML features, selected approaches, and built and trained models.

**Earlier Data Science, NLP & Software Roles** - Nestlogic, Maverick Medical AI, Athena Portfolio Solutions, Ten-X, Verba, and Finfon 2009-2019

Roles in Tel Aviv, Hong Kong, and Prague.

- Financial-news NLP and signal extraction; medical-text classification and information extraction; computer-vision pipelines; and corpus-linguistics infrastructure.

### SELECTED WORK (PRIVATE REPOSITORIES, CODE ON REQUEST)

**pi-harness** - Eval harness for a coding agent: probe suite scoring pass/fail, cost, and wall time per run. An A/B of a structural gate moved spec-first compliance from 0/3 to 3/3 at +26% cost and +80% wall time (n=3, one model).

**inference_harness** - Elixir LLM harness for multi-provider routing, role-specific fallback, token/latency tracing, and bounded concurrent jobs.

**gemini-gate** - Elixir/Phoenix credential service with authenticated retrieval, Gemini API proxying, and Caddy/systemd deployment.

### BACKGROUND

**Education:** MSc Computational Linguistics, RSUH, Moscow, 2015. BSc Nuclear Physics, Czech Technical University, Prague, 2009.

**Languages:** English, Hebrew, Russian; working knowledge of German and French.

Artifact provenance: project_name=marchdown.github.io; repo_path=/Users/uprootiny/marchdown.github.io/.claude/worktrees/cv-salestech-ai; git_remote=https://github.com/marchdown/marchdown.github.io.git; git_branch=worktree-cv-salestech-ai; git_commit=ac7e228eb93c2896b43d6c745eb62e65694e7db0.

generated_at=2026-10-25T13:08:19.613Z; generator=Codex (GPT-5) via scripts/generate-cv.mjs (Node 22.23.0); session_source=Codex API conversation; session_locator=/Users/uprootiny/.codex/sessions/ (2026-07-18 session).